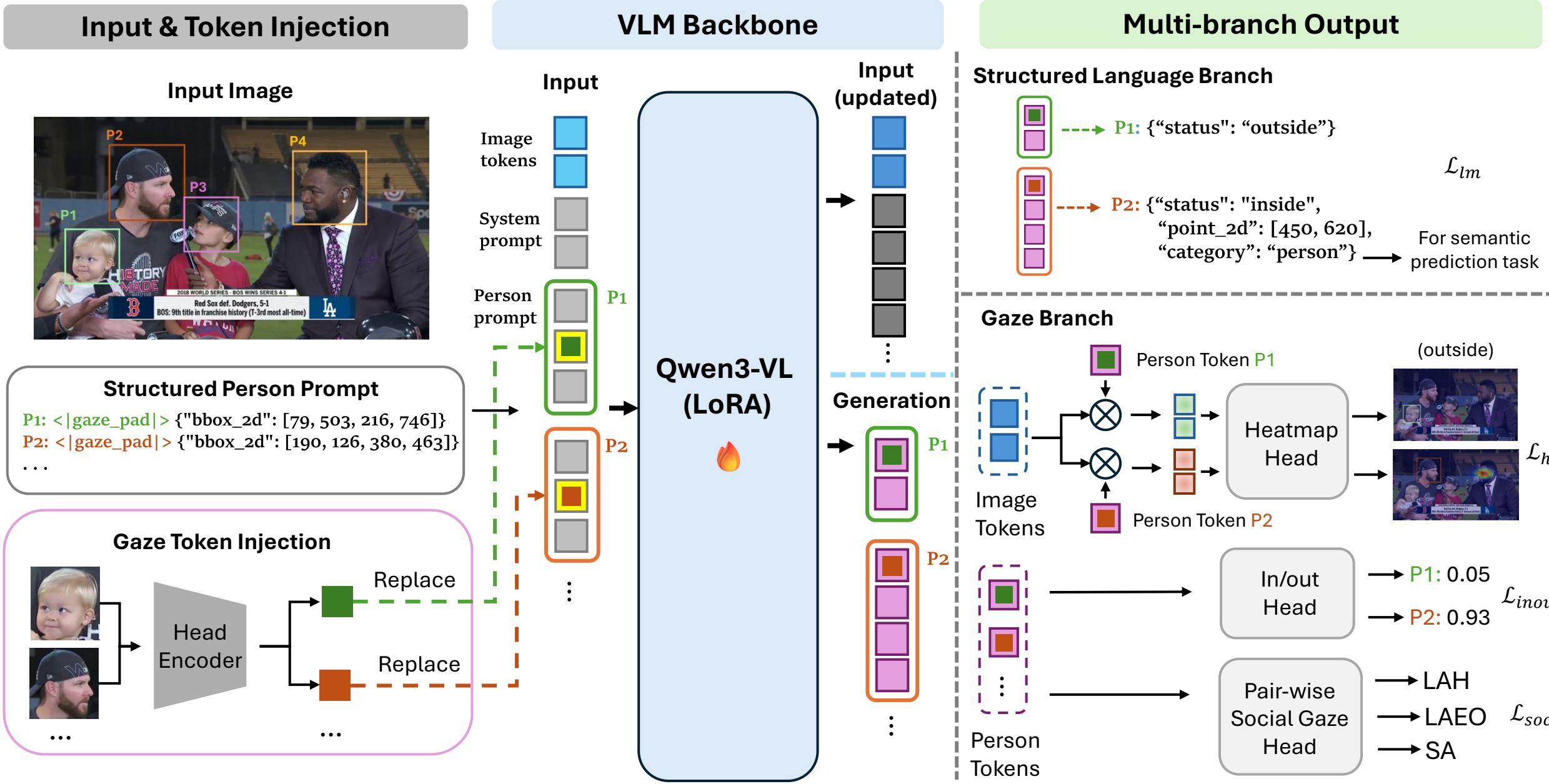

Input & Token Injection
VLM Backbone
Multi-branch Output
Input Image
P1
P2
P3
P4
Structured Person Prompt
P1: <|gaze_pad|> {"bbox_2d": [79, 503, 216, 746]}
P2: <|gaze_pad|> {"bbox_2d": [190, 126, 380, 463]}
. . .
Gaze Token Injection
Head Encoder
Replace
Replace
Input
Image tokens
System prompt
Person prompt
P1
P2
Qwen3-VL (LoRA)
Input (updated)
Generation
P1
P2
Structured Language Branch
P1: {"status": "outside"}
P2: {"status": "inside", "point_2d": [450, 620], "category": "person"}
For semantic prediction task
$\mathcal{L}_{lm}$
Gaze Branch
Person Token P1
Person Token P2
Image Tokens
Heatmap Head
(outside)
$\mathcal{L}_{hm}$
Person Tokens
In/out Head
P1: 0.05
P2: 0.93
$\mathcal{L}_{inout}$
Pair-wise Social Gaze Head
LAH
LAEO
SA
$\mathcal{L}_{soc}$
Image Tokens
Prompt Tokens
<gaze_pad> Tokens
Hidden States for generated prompts